



DATA SCIENCE



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Today's Agenda

- Review of previous lecture
- How to Do Data Science?
- Languages, Tools and Techniques
- Life Cycle of a Data Science Project
- Industry Job Roles in Data Science



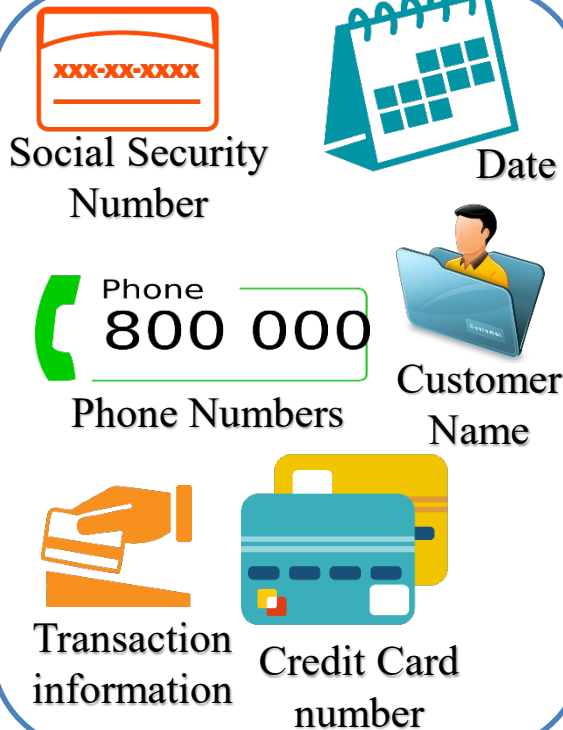


Recap of Previous Lecture



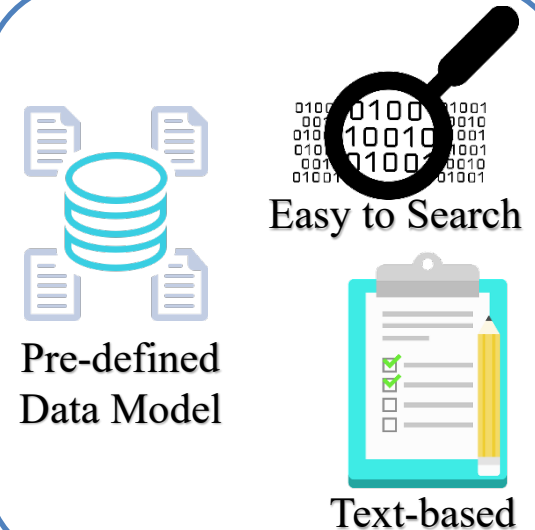
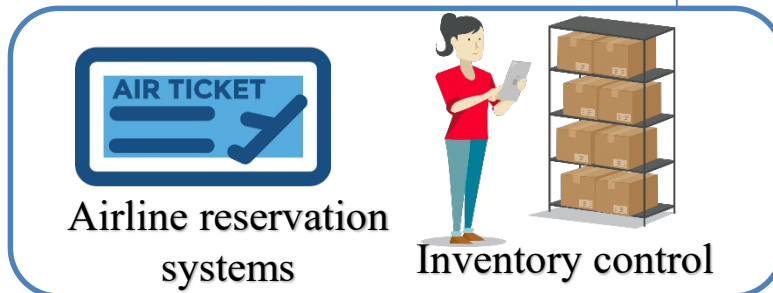
Structured Data

Examples

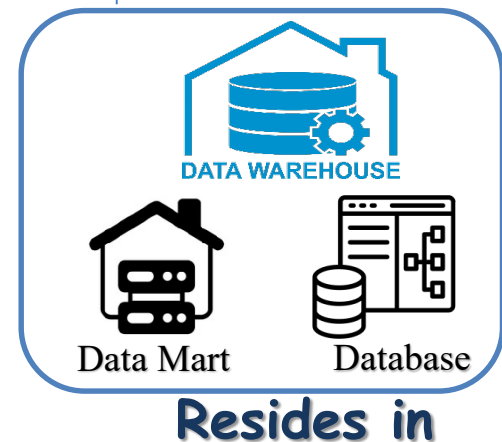


Structured Data

Applications



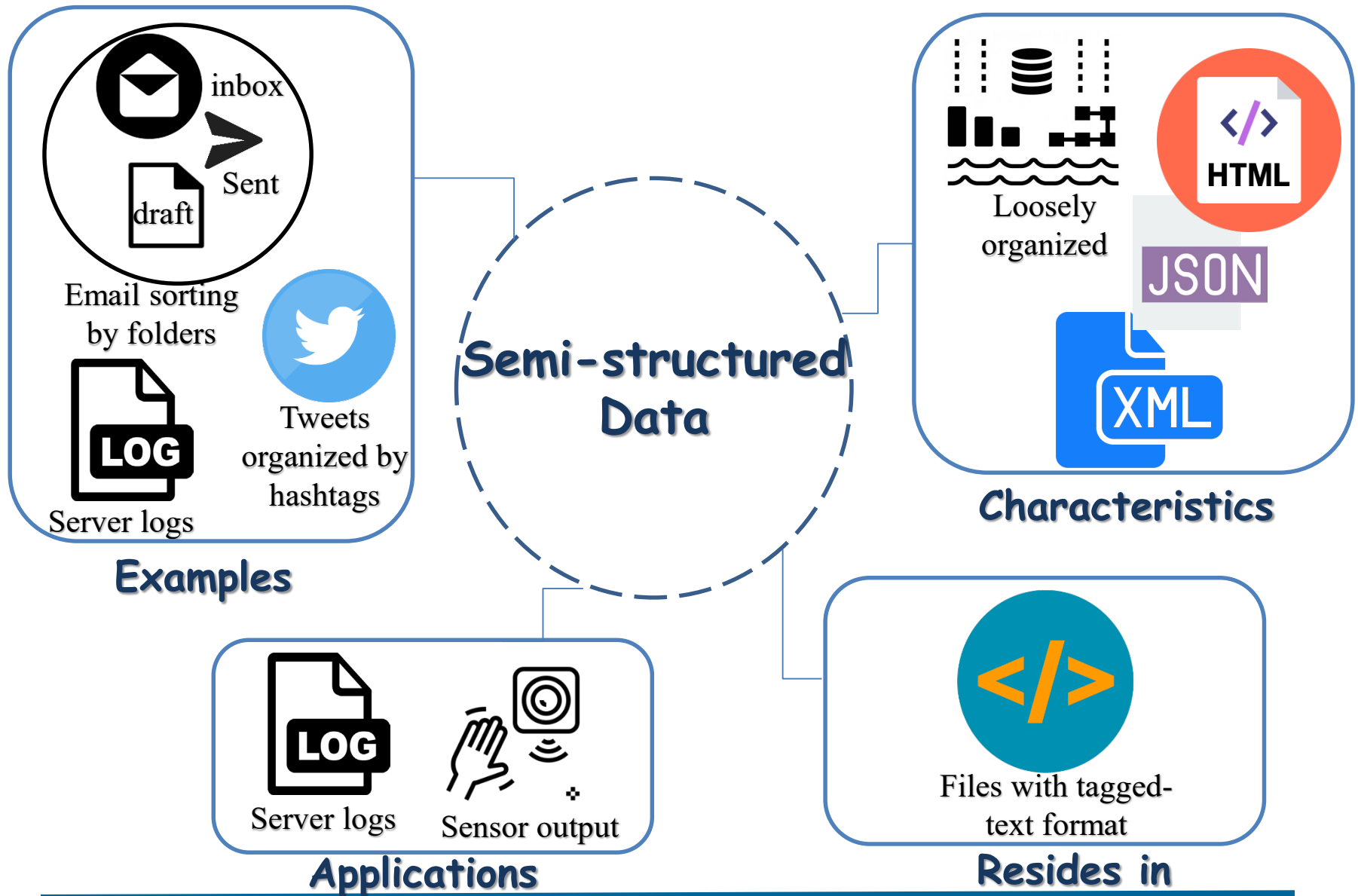
Characteristics



Resides in

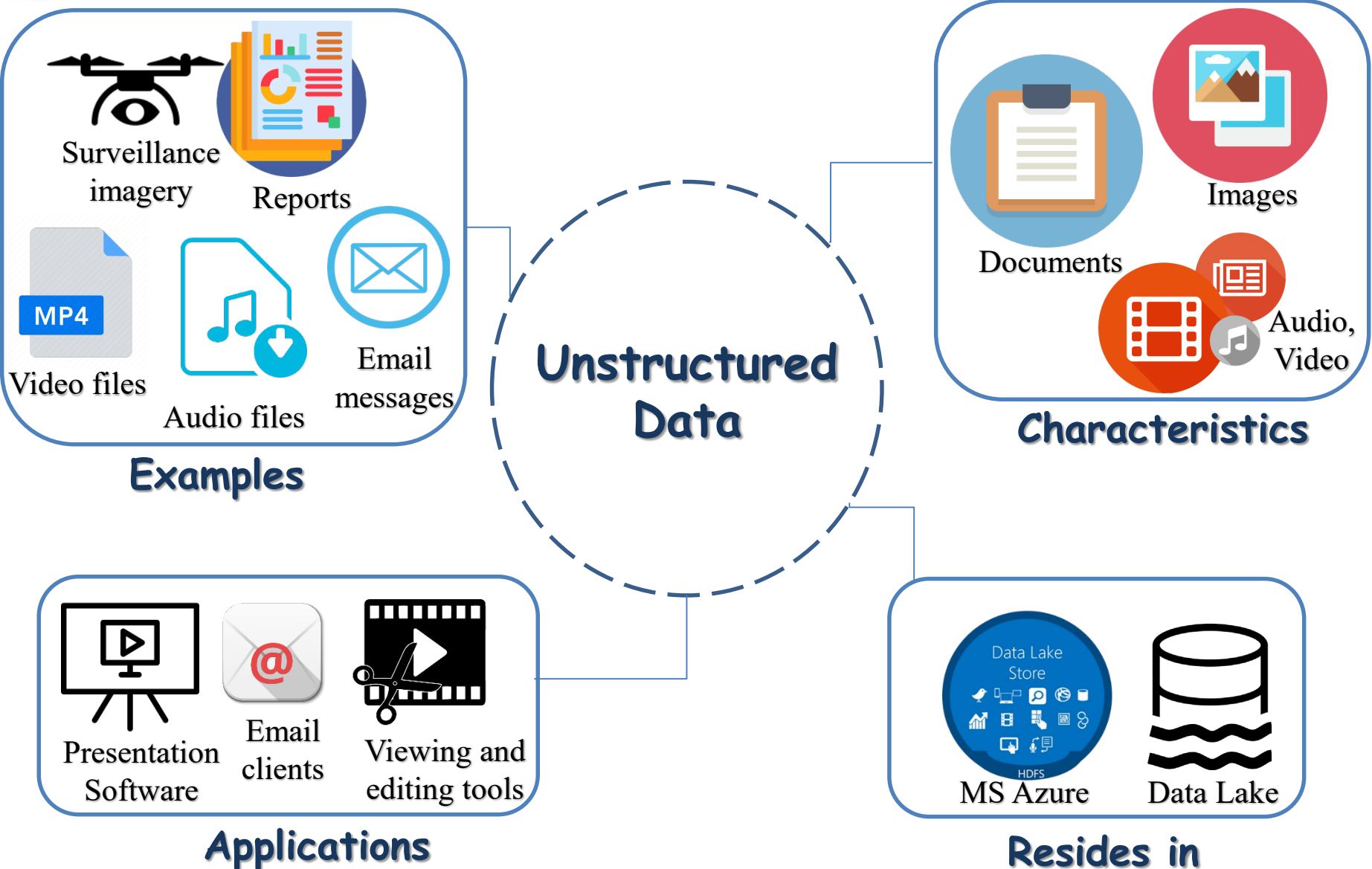


Semi-structured Data





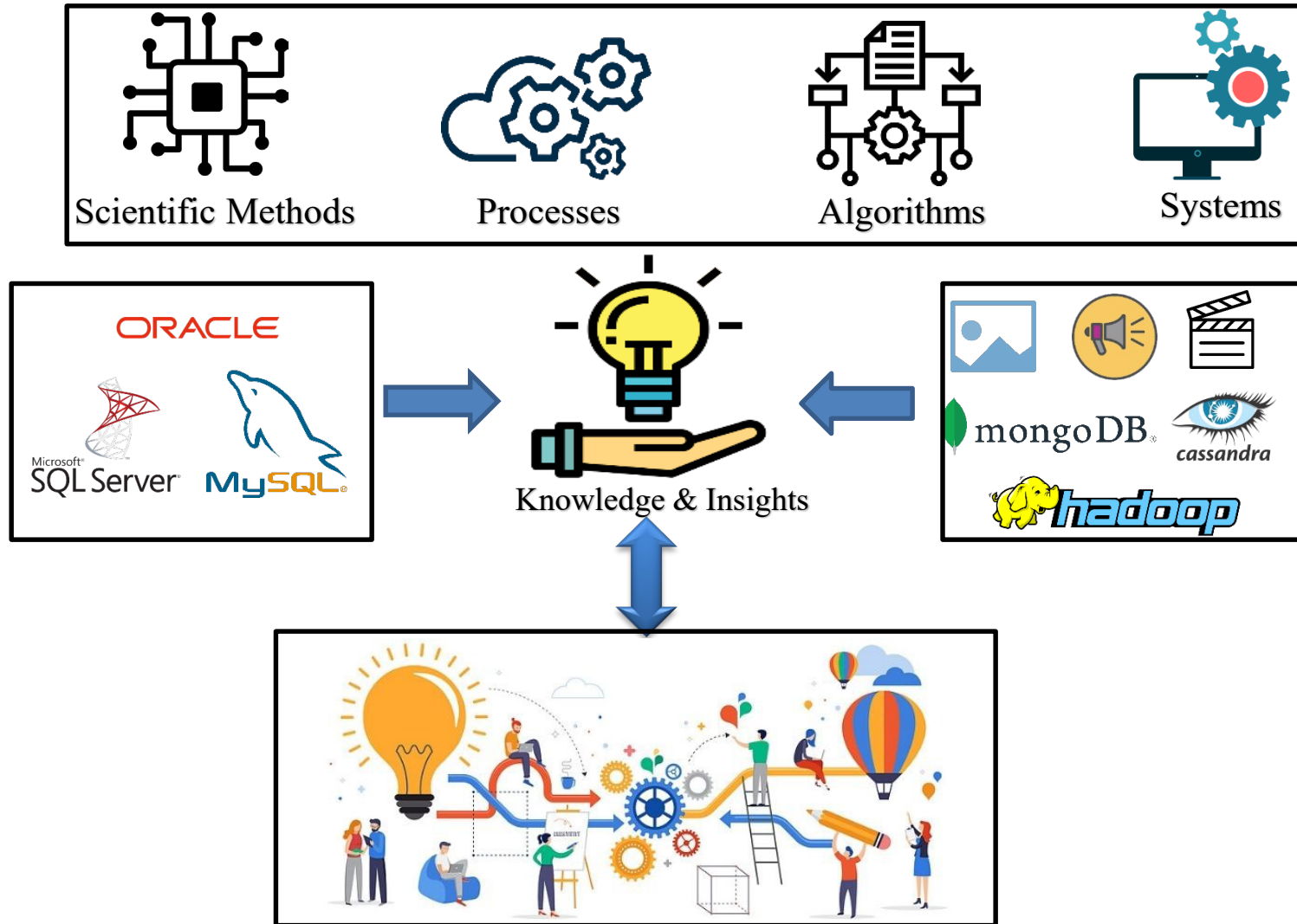
Unstructured Data





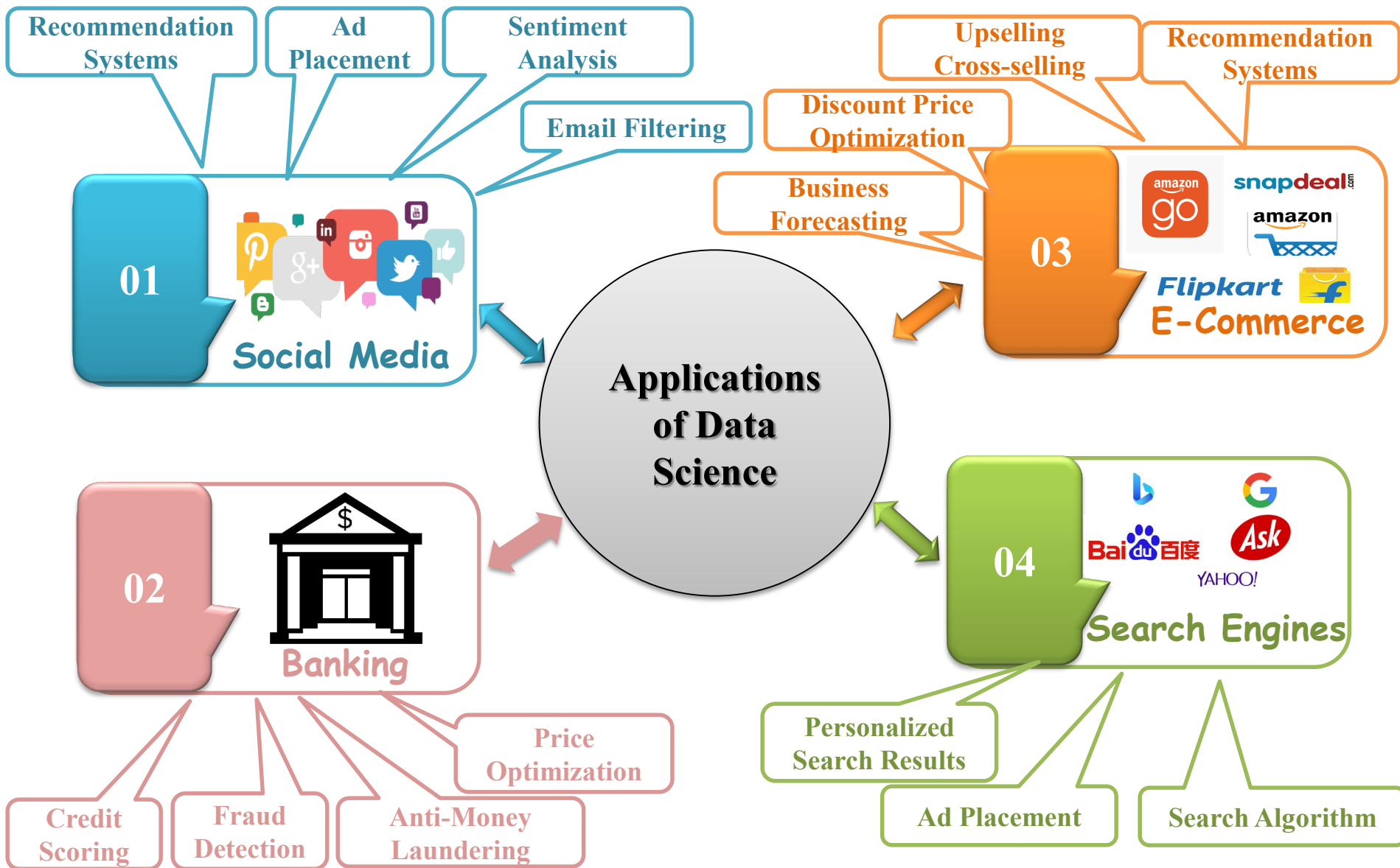
What is Data Science?

Data Science is an Inter-Disciplinary Field that uses



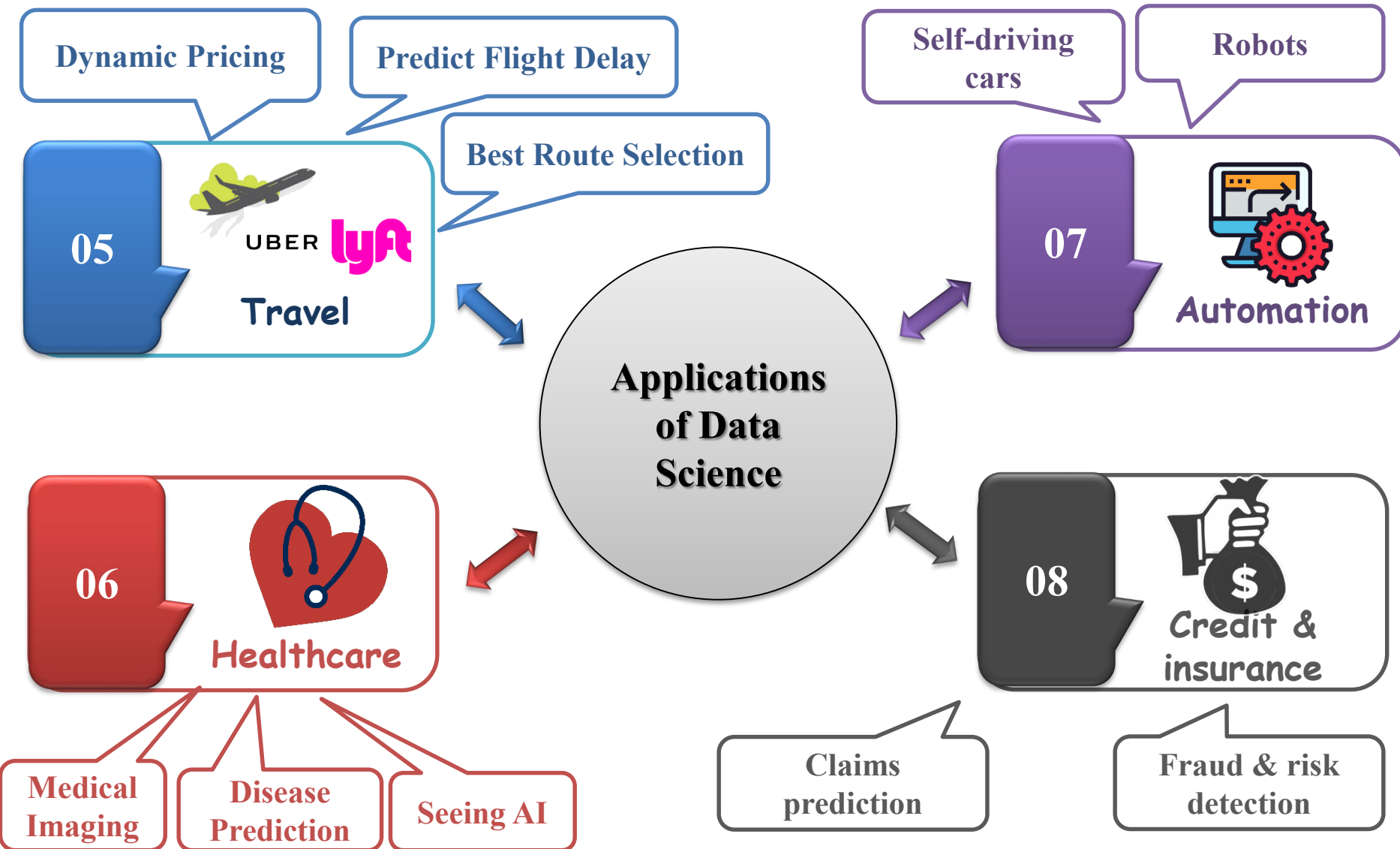


Applications of Data Science





Applications of Data Science (cont...)





How to Do Data Science?

Languages, Tools and Technologies



Who is a Data Scientist?

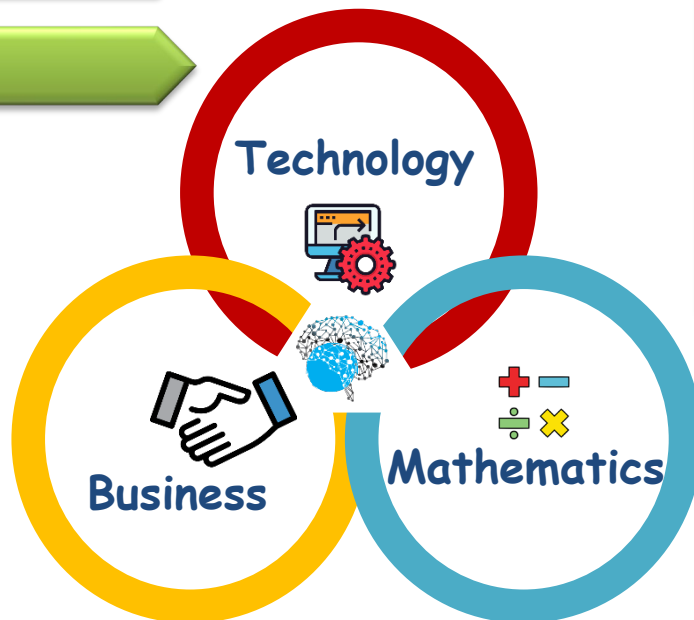
1 Data Scientist

2 Skill Set

3 Programming languages

4 Tools

5 Techniques



A data scientist is a professional responsible for **collecting, analyzing** and **interpreting** extremely large amounts of structured and unstructured data in order to gain useful insights to grow the business



Skill Sets of a Data Scientist

1

Data Scientist

2

Skill Set

3

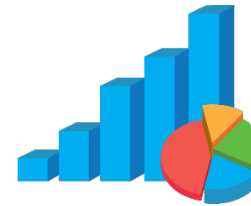
Programming languages

4

Tools

5

Techniques



Statistics



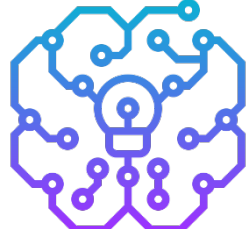
Programming
Languages



Data extraction
& processing



Big Data processing
framework



Machine Learning



Data
Visualization



Data wrangling
& exploration



Programming Languages for Data Science

1 Data Scientist

2 Skill Set

3 Programming language

4 Tools

5 Techniques



Python



R



Julia





Tools for Handling this Big Data (3Vs)

Tools are softwares that are used to apply DS techniques to perform a task.



VOLUME



VARIETY



VELOCITY



Python Libraries for Data Science Tasks



SciPy



pandas



SQLAlchemy

Scrapy



matplotlib

seaborn



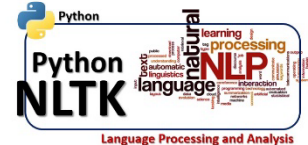
Keras



PyTorch



semantria





Techniques for Data Science

1 Data Scientist

2 Skill Set

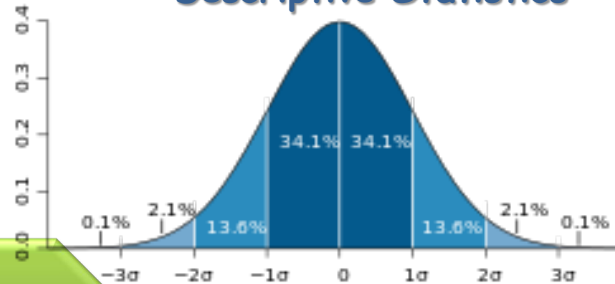
3 Programming language

4 Tools

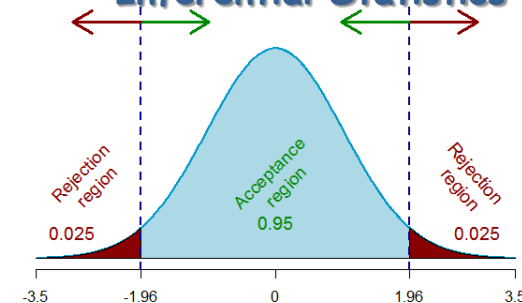
5 Techniques

Techniques are set of procedures that are followed to perform a task. Tools and techniques together helps in data collection, data storage, data preparation, data analysis, data modeling and data visualization

Descriptive Statistics

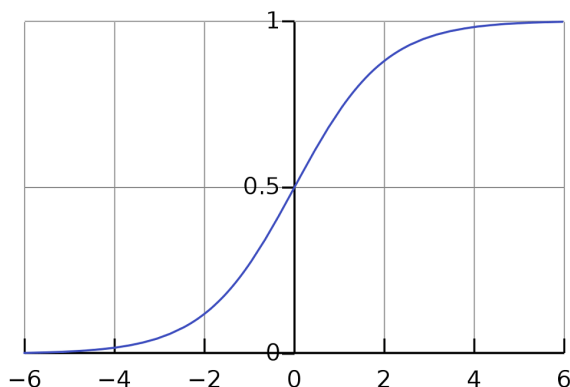


Inferential Statistics

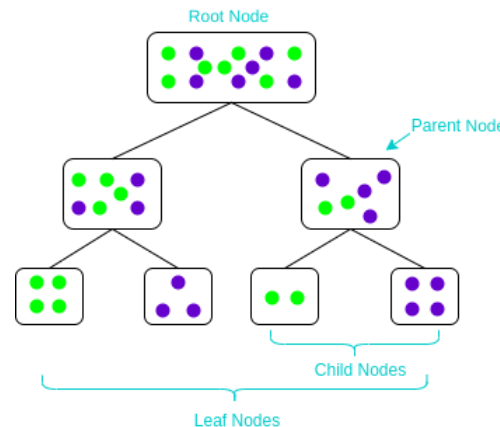


Classification Techniques

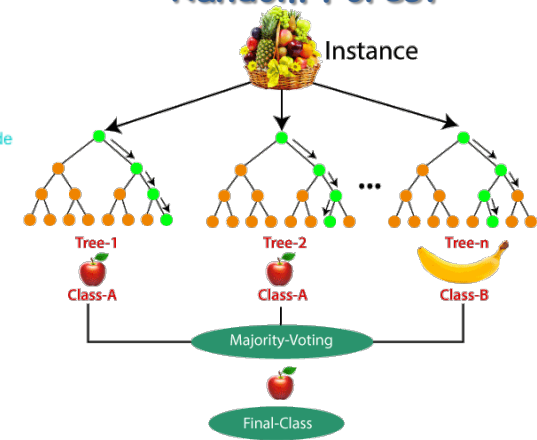
Logistic Regression



Decision Tree



Random Forest





Techniques for Data Science

1 Data Scientist

2 Skill Set

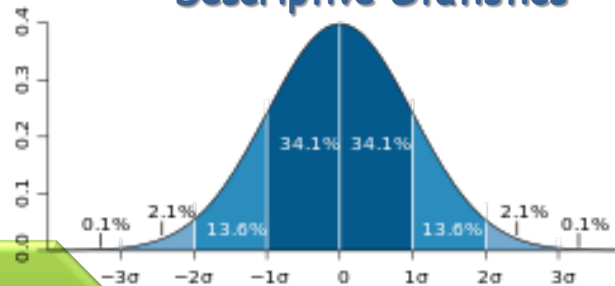
3 Programming language

4 Tools

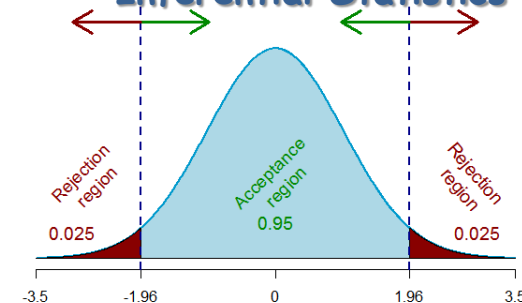
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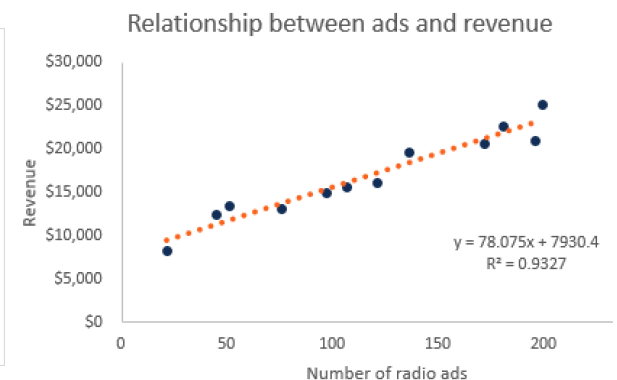
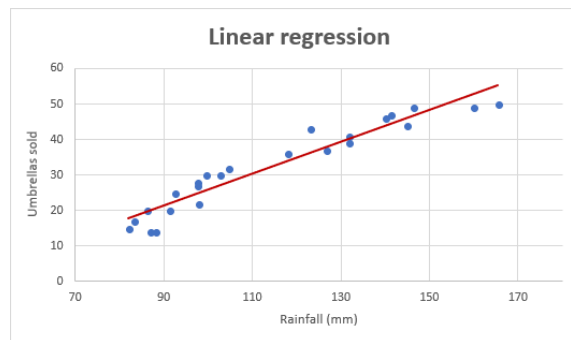
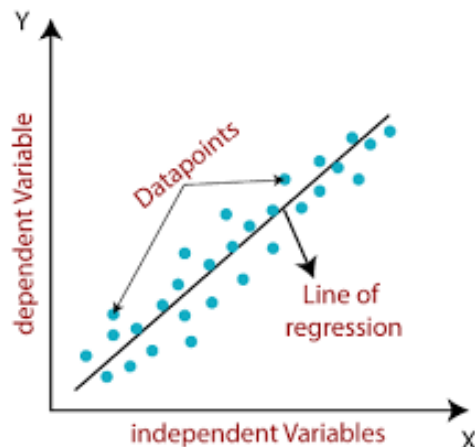
Descriptive Statistics



Inferential Statistics



Regression Techniques





Techniques for Data Science

1 Data Scientist

2 Skill Set

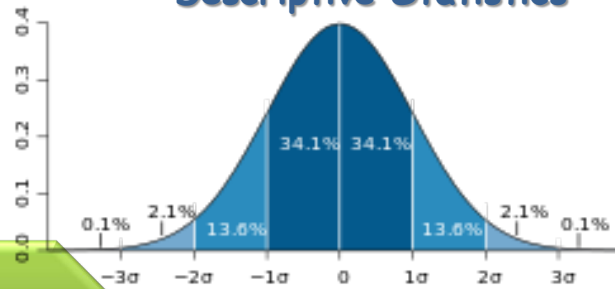
3 Programming language

4 Tools

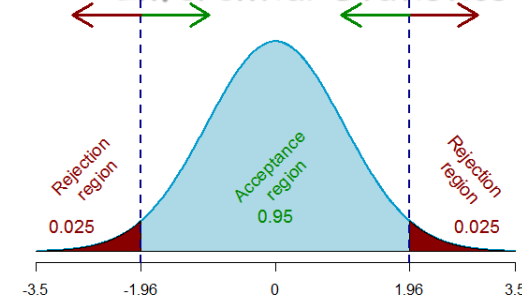
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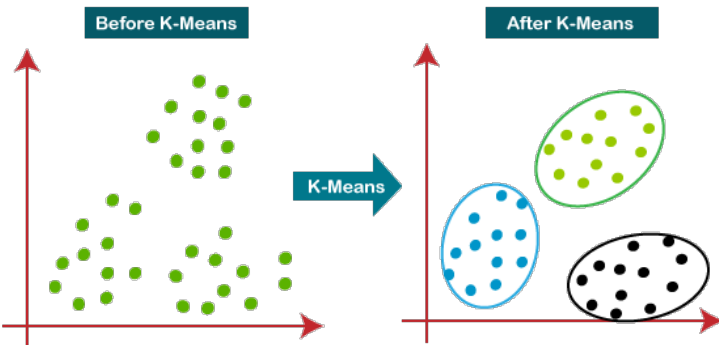


Inferential Statistics

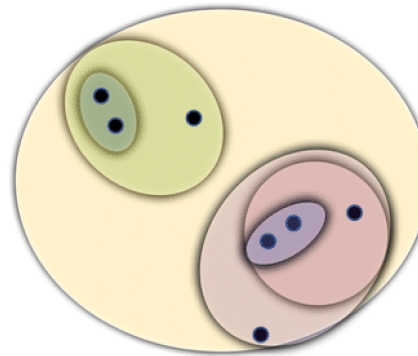


Clustering Techniques

K-Means Clustering



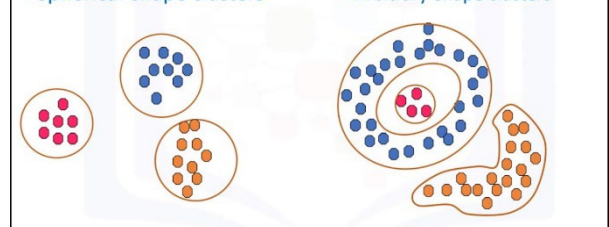
Hierarchical Clustering



DB SCAN

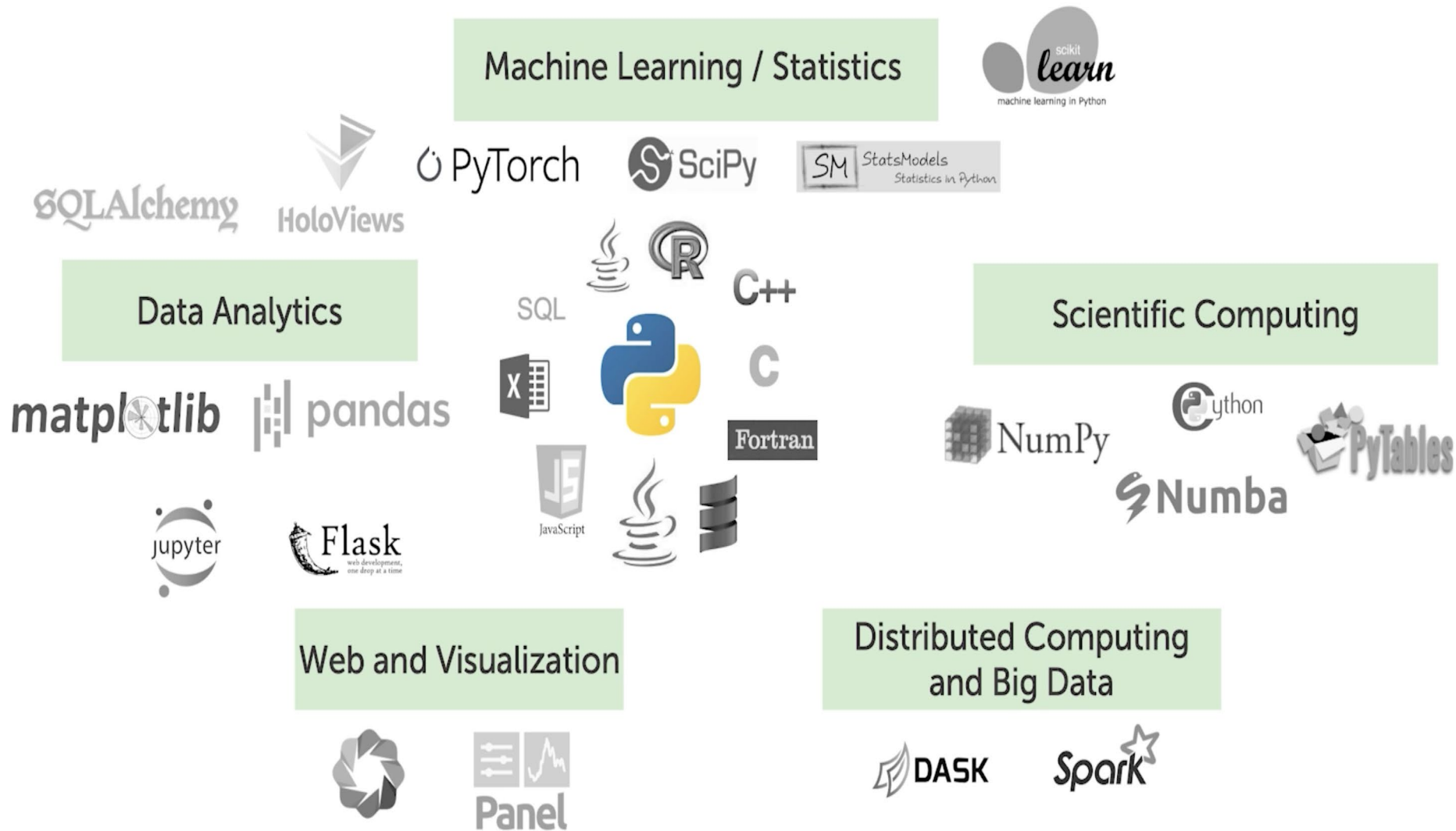
Density-based clustering

- Spherical-shape clusters
- Arbitrary-shape clusters





Why is Data Science so Complicated?

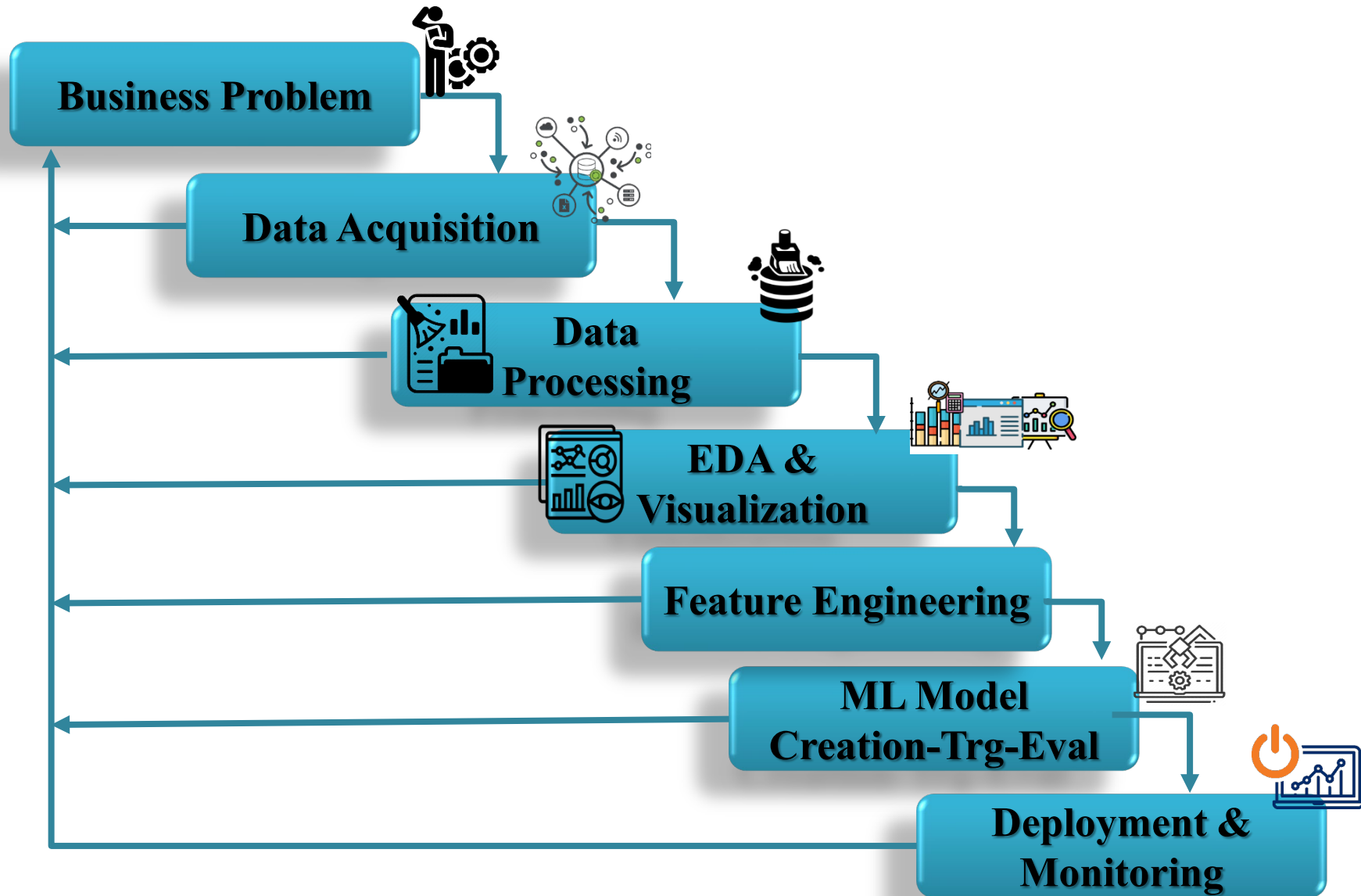




Data Science Life Cycle

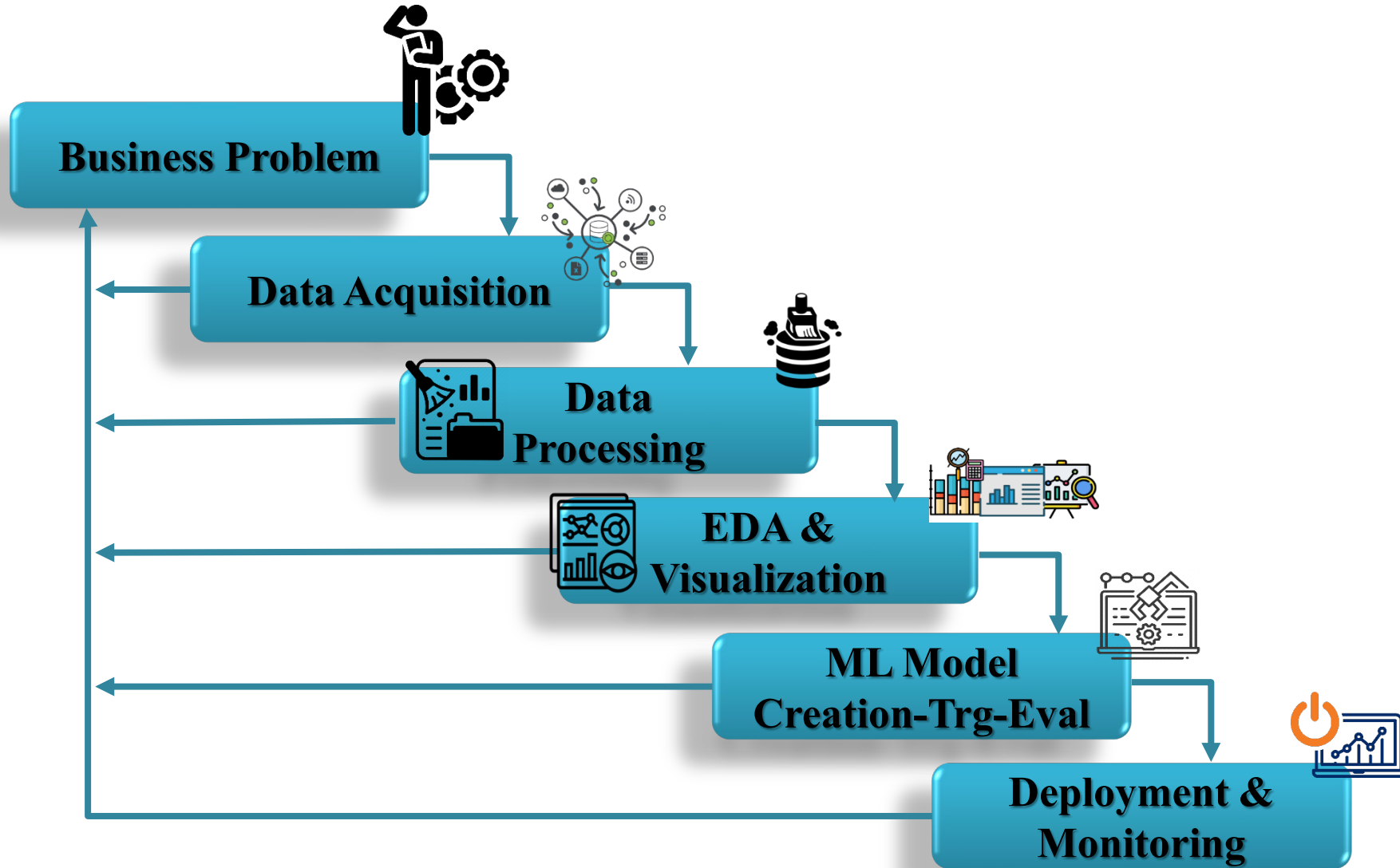


Overview of Data Science Life Cycle





Overview of Data Science Life Cycle





Understanding Business Problem

1 Business Problem

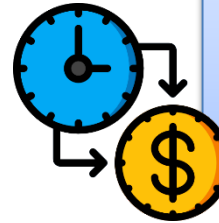
2 Data Acquisition

3 Data Processing

4 EDA & Visualization

5 Model Creation-Trg-Eval

6 Deployment & Monitoring



Most critical phase of a Data Science Life Cycle, if conducted will save a lot of time, money and resources.

Understand the problem by talking to the stakeholders & domain experts to get a clear understanding of the problem and document all the requirements.

WHY?...WHY?...WHY?...



Identify the key business variables that need to be predicted
Define the success criteria and success measuring metrics (KPIs & SLAs)



Data Acquisition

1 Business Problem

What data do we need for our project?



2 Data Acquisition

3 Data Processing

4 EDA & Visualization

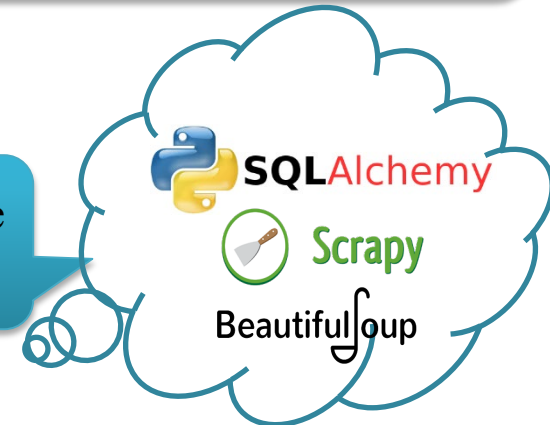
5 Model Creation-Trg-Eval

6 Deployment & Monitoring

What are the data sources and data format?
Where is the data located?



How can we obtain the data?

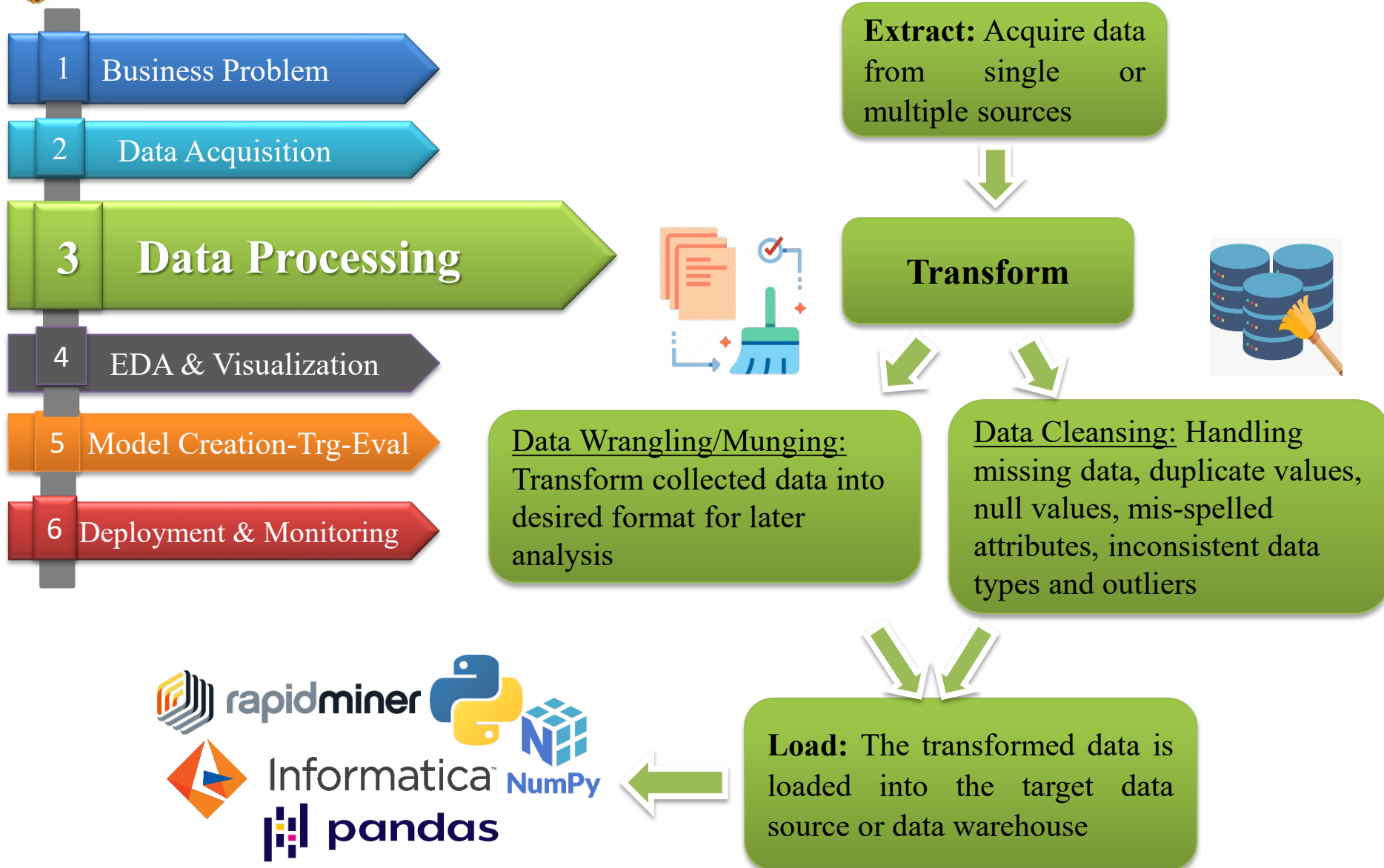


What is the most efficient way to store and access all of it for later processing?



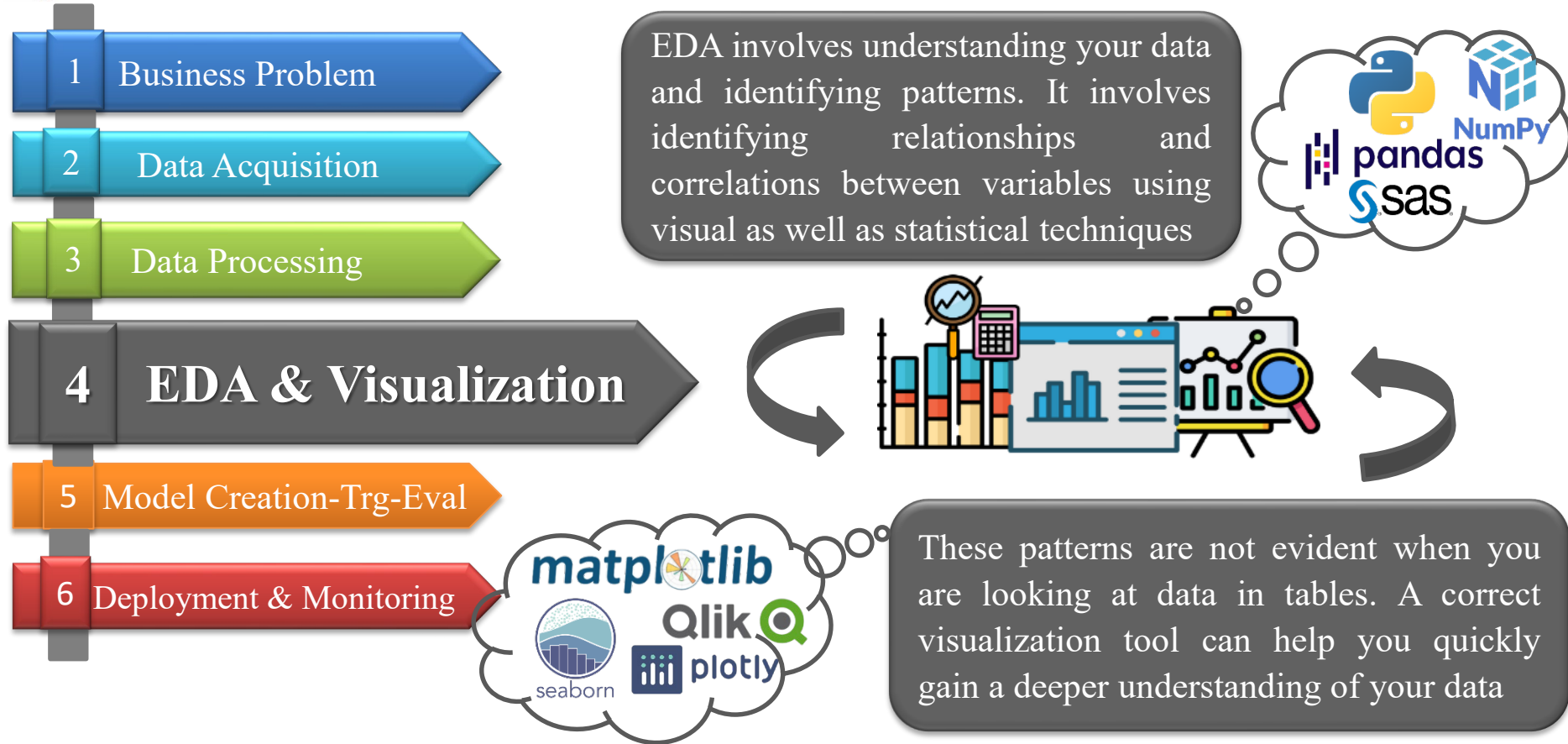


Data Processing





Exploratory Data Analysis & Visualization



Finally EDA involves Feature Engineering, which performs feature creation, transformation, extraction and selection before creation of ML model

Data Analyst's Job Ends Here



ML Model: Creation-Training-Evaluation

1 Business Problem

2 Data Acquisition

3 Data Processing

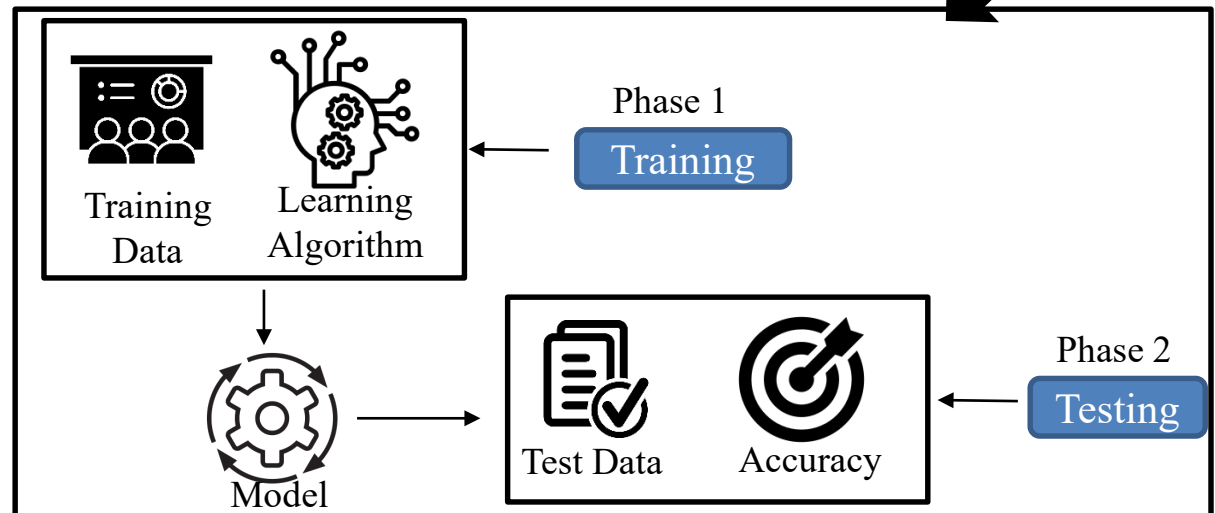
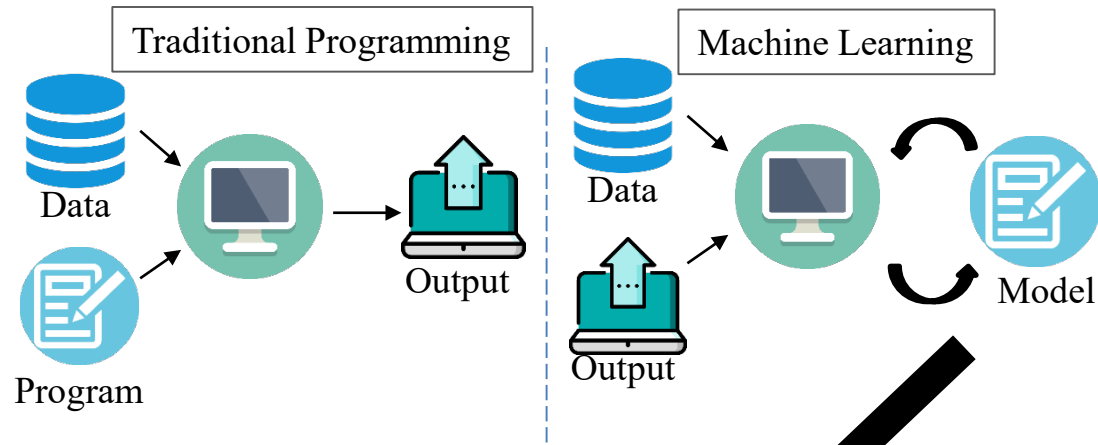
4 EDA & Visualization

5 **ML Model
Creation-Trg-Eval**

6 Deployment & Monitoring

Use different but appropriate machine learning algorithms like Decision Tree, Linear Regression, K-Nearest Neighbour to the data to identify the model that best fits the business requirements

ML is an application of AI that gives computers the ability to learn without being explicitly programmed. [Arthur Samuel]





Model Deployment and Monitoring

1 Business Problem

2 Data Acquisition

3 Data Processing

4 EDA & Visualization

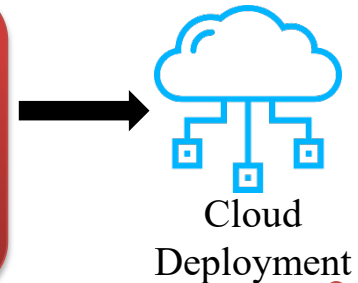
5 Model Creation-Trg-Eval

6 **Deployment & Monitoring**

After a model is trained, tuned and tested, you can deploy the model into production and make inferences (predictions)



Check the deployment environment for dependency issues
Deploy the model first in the test and then in the production environment



Most of the times the live real world data differ from the data that was used to train the model, thus making the model less accurate. To handle this, build a model monitor that detects deviations such as data drift and alerts you to take remedial actions





Industry Job Roles in Data Science



Industry Job Roles: Data Scientist

1 Data Scientist

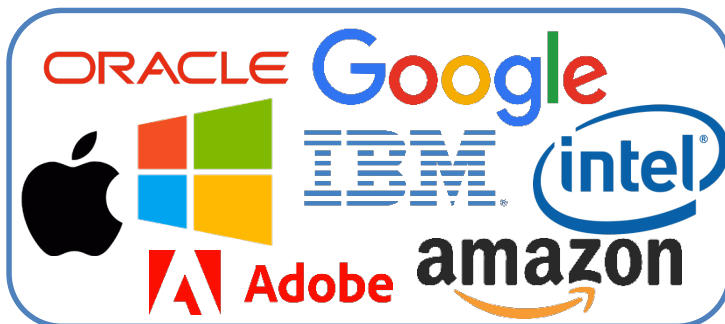
- Senior most in the team and take inputs from the rest to formulate actionable insight for the business
- Makes use of the latest tools and technologies in finding solutions and reaching conclusions that are crucial for an organization's growth and development

2 Data Engineer

3 Data Analyst

4 Database Administrator

5 ML Engineer





Industry Job Roles: Data Engineer/Architect

1

Data Scientist

2

Data Engineer

3

Data Analyst

4

Database Administrator

5

ML Engineer

- Scrape data and store it in warehouses using ETL
- Handle databases and create data warehouses
- Design, build, and manage the big data infrastructure
- Build data pipelines for easy access of data
- Big Data Tools (Apache Spark, Apache Hive, Hadoop)
- Cloud Platforms (AWS, Google Cloud Platform)





Industry Job Roles: Data Analyst

1

Data Scientist

2

Data Engineer

3

Data Analyst

4

Database Administrator

5

ML Engineer

- Data Analyst is an entry level member into the data analytics team
- Needs to have good technical skills and know the basics of statistics, data munging, data utilization, and exploratory data analysis
- Generate reports after analyzing the data
- Can move to the role of Data engineer and Data scientist with more experience





Industry Job Roles: Database Administrator

1 Data Scientist

2 Data Engineer

3 Data Analyst

4 **Database Administrator**

5 ML Engineer

- Responsible for administering the collected data by installing, configuring, monitoring, operating, and maintaining database
- Ensure that all databases are available to all relevant users, and is protected securely from any malicious activity





Industry Job Roles: Machine Learning Engineer

1 Data Scientist

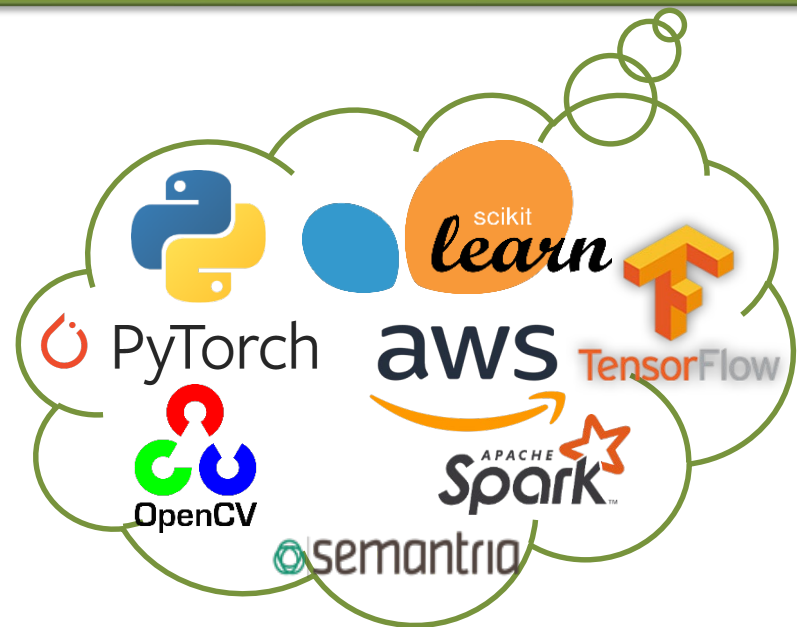
2 Data Engineer

3 Data Analyst

4 Database Administrator

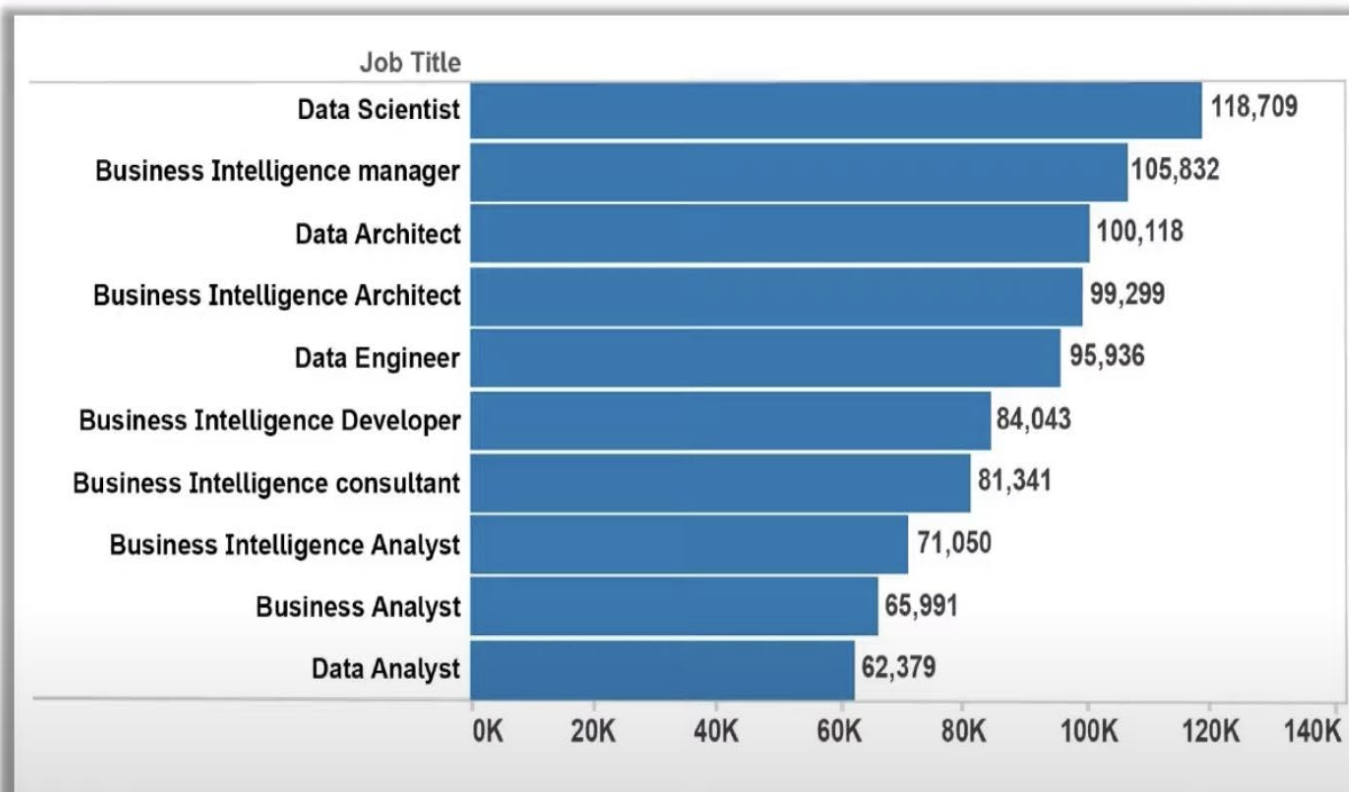
5 ML Engineer

- Machine learning engineer works as a part of large data science team
- Responsible to design and create all algorithms capable of learning and making predictions
- They are expected to perform A/B testing, build data pipelines, and implement algorithms for classification, clustering, regression, anomaly detection etc.





History: Data Science Salary Trends



National average salary for different job roles in data science

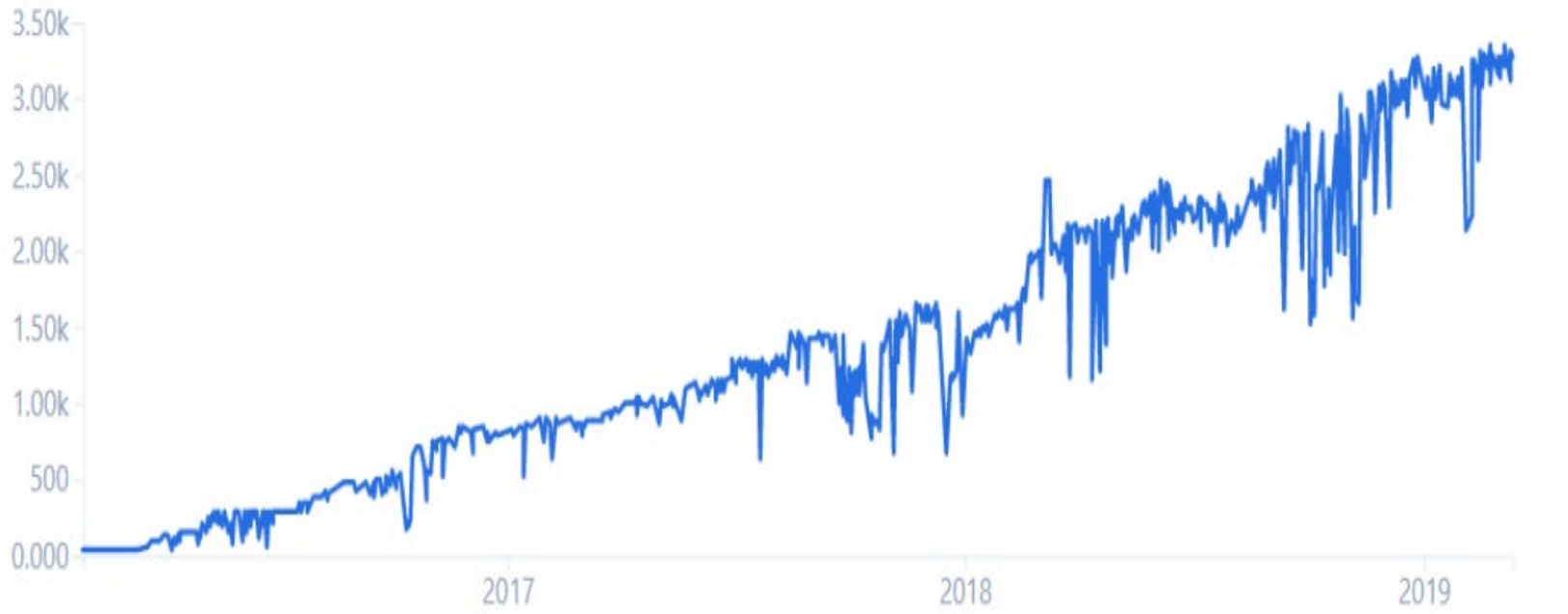
Source - Glassdoor

Source: <https://towardsdatascience.com/why-learn-data-science-in-2020-d3f54123b2e4>



History: Job trends

Data Scientist job openings at the world's top companies



Data from Thinknum - Open dataset

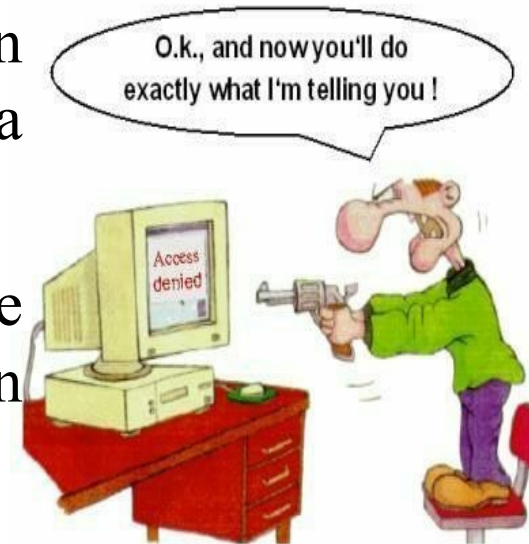
● Title (Count)

Source: <https://www.tecla.io/blog/the-high-demand-for-data-scientists-and-how-to-hire-for-them/>



Things To Do

- Visit all the hyperlinked tools and technologies in today's lecture slides. You should be able to give a single line description of each.
- Have a very clear understanding of Data Science Life Cycle, the tools & the technologies used in each phase.
- Think of few use cases where you can apply Data Science, Machine Learning and Deep Learning technologies and make a list of the skill set you need to develop/learn to implement and deploy such projects.



Coming to office hours does NOT mean you are academically weak!